

JUNE, 1901

A SHORT LIST

OF THE

PUBLICATIONS

OF

E. & F. N. SPON, Ltd.,

125, STRAND, LONDON, W.C.

JUNE, 1901

A SHORT LIST

OF THE

PUBLICATIONS

OF

E. & F. N. SPON, Ltd.,

125, STRAND, LONDON, W.C.

The Project Gutenberg eBook of A Short List of the Publications of E. & F. N. Spon, Ltd. June 1901

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: A Short List of the Publications of E. & F. N. Spon, Ltd. June 1901

Author: E. & F. N. Spon

Release date: November 14, 2018 [eBook #58286]

Language: English

Other information and formats: www.gutenberg.org/ebooks/58286

Credits: Produced by Chris Curnow, Chris Jordan and the Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK A SHORT LIST
OF THE PUBLICATIONS OF E. & F. N. SPON, LTD. JUNE 1901 ***

JUNE, 1901

**A SHORT LIST
OF THE
PUBLICATIONS
OF
E. & F. N. SPON, Ltd.,
125, STRAND, LONDON, W.C.**

Full particulars post free on application.
All books are bound in cloth unless otherwise stated.

AGRICULTURE.

Brooks, C. P. Cotton: its uses, varieties, Fibre Structure, Cultivation and Preparation for the Market, and as an Article of Commerce, 8vo <i>net</i>	7	6
Lock, C. G. W. Coffee: its Culture and Commerce in all Countries. Crown 8vo	12	6
—— Tobacco Growing, Curing and Manufacturing. Crown 8vo	6	0
Lock, C. G. W., & Newlands Bros. Sugar: a Handbook for Planters and Refiners, 8vo	1	10 0
Simmonds, P. L. Hops: their Cultivation, Commerce and Uses. Crown 8vo	4	6
—— Tropical Agriculture. New edition, 8vo	1	1 0

ARCHITECTURE AND BUILDING.

Allsop, R. O. The Hydropathic Establishment and its Baths. Demy 8vo	5	0
—— The Turkish Bath, its Design and Construction. Demy 8vo	6	0
—— Public Baths and Wash Houses. Demy 8vo	6	0
Bales, T. The Builder's Clerk. Second edition. fcap. 8vo	1	6
Burns, G. J. Glossary of Technical Terms used in Architecture and the Building Trades. Crown 8vo	3	6
Christie, W. W. Chimney Design and Theory. Crown 8vo <i>net</i>	12	6

Coleman, T. E. Stable Sanitation and Construction. Crown 8vo	6	0	
—— Architect's Price Book. 32mo, roan, gilt edges			
Fullerton, W. Architectural Examples in Brick, Stone, Wood and Iron. Second edition, demy 4to	15	0	
Hoskins, G. G. The Clerk of Works' Vade Mecum. Seventh edition, fcap. 8vo	1	6	
Hurst, J. T. A Handbook of Formulæ, Tables and Memoranda for Architectural Surveyors, and others engaged in Building. Fifteenth edition, royal 32mo, roan	5	0	
Laxton. Builders' and Contractors' Tables. Excavator, Earth, Land, Water and Gas. 4to	5	0	
Leaning, J. A complete set of Contract Documents for a Country Lodge, Printed in facsimile of the original. fcap., in case	5	0	
—— Quantity Surveying. Fourth edition, 8vo	15	0	
Macey, F. W. Specifications in Detail. Medium 8vo	1	1	0
Piranesi. Roman Architecture, Sculpture and Ornament. 200 plates, reproduced in facsimile from the original. Imperial folio, half morocco	5	5	0

Sharpe, E. The Seven Periods of English Architecture, defined and illustrated. Third edition, royal 8vo	12	6
Spons' Architects' and Builders' Price Book. By W. YOUNG. Edited by CLYDE YOUNG. 8vo, published annually	3	6
Thwaite, B. H. Our Factories, Workshops and Warehouses, their Sanitary and Fire-Resisting Arrangements. Crown 8vo	9	0
Tredgold, T., & Hurst, J. T. Elementary Principles of Carpentry. Tenth edition, crown 8vo	12	6
Wood, W. H. Practical Stair Building and Handrailing. Crown 4to	10	6
Young, W. The Municipal Buildings, Glasgow. Folio	7	6

BRIDGES, ROOFS, ETC.

Adams, Hy. Strains in Ironwork. Third edition, crown 8vo	5	0
—— The Practical Designing of Structural Ironwork. 8vo	8	6
Bell, G. J. A Practical Treatise on Segmental and Elliptical Oblique or Skew Arches. Royal 8vo <i>net</i>	1	1 0
Bow, R. H. Economics of Construction in relation to Framed Structures. Third thousand. 8vo	5	0

Donaldson, W. The Art of Constructing Oblique Arches with Spiral Courses. 8vo	4	0	
—— New Formulæ for the Loads and Deflections of Solid Beams and Girders. Second edition, 8vo	4	6	
Fitzmaurice, M. Plate Girder Railway Bridges. 8vo	6	0	
George, E. M. Pocket Book of Calculations in Stresses. Royal 32mo, half roan	3	6	
Grover, J. W. Estimates and Diagrams of Railway Bridges for Turnpike, Public and Occupation Roads, with Form for Calculating Quantities in Skew Structures, etc. Second edition, folio	1	11	6
—— Iron and Timber Railway Superstructures and General Works. Folio	2	2	0
Heaford, A. S. Strains on Braced Iron Arches and Arched Iron Bridges. 8vo	6	0	
Johnson, F. R. Stresses in Girder and Roof Frames for both dead and live loads by simple Multiplication, etc. Crown 8vo	6	0	
Newman, J. Notes on Cylinder Bridge Piers and the Well System of Foundations, 8vo	6	0	
Olander, E. A New Method of Graphic Statics applied in the Construction of Wrought Iron Girders. Small folio	10	6	

Ritter, A. Elementary Theory and
Calculation of Iron Bridges and Roofs.
Translated from the Third German
Edition by H. R. SANKEY. 8vo 15 0

Stoney, B. B. The Strength and Proportion
of Rivetted Joints. 8vo 5 0

Walmisley, A. T. Examples of Iron Roofs.
Small folio, half Morocco. *Third
Edition in preparation*

**CHEMISTRY, see INDUSTRIAL CHEMISTRY. CIVIL
ENGINEERING.**

IRRIGATION, SURVEYING, WATERWORKS.

Allen, J. R. Theory and Practice in the
Design and Construction of Dock
Walls. Crown 4to 6 0

Bligh, W. G. Notes on Instruments best
suited for Engineering Field Work in
India and the Colonies. 8vo 7 6

Buckley, R. B. Irrigation Works in India
and Egypt. Imperial 8vo 3 3 0

Codd, W. Land Area Tables. Square 16mo,
on a sheet mounted on linen and bound
in cloth 3 6

Codrington, T. The Maintenance of
Macadamised Roads. Second Edition,
8vo 7 6

Cordeiro, F. J. B. The Barometrical
Determination of Heights. Crown 8vo,
limp leather 4 6

Coventry, W. B. On Curved Masonry Dams. 8vo sewed	2	0
—— A Practical Method of Determining the Profile of a Masonry Dam. 8vo, sewed	2	0
—— The Stresses on Masonry Dams (oblique sections), 8vo, sewed	2	6
Cunningham, D. Tables for facilitating the Calculation of Earthworks in the construction of Railways, Roads, Canals, Dams, etc. Royal 8vo	10	6
Cutler, H. A., & Edge, F. J. Tables for Setting out Curves from 101 to 5000 feet radius. Royal 32mo	2	6
De Salis, H. R. A Chronology of Inland Navigation in Great Britain. Crown 8vo	4	6
Gripper, C. Railway Tunnelling in Heavy Ground. Royal 8vo	7	6
Holloway, T. Levelling and its General Application. Second Edition, 8vo	5	0
Jeanes, J. S. Waterways and Water Transport in different Countries. 8vo	14	0
Matheson, E. Aid Book to Engineering Enterprise. Third edition, medium 8vo, buckram	1	4 0
Matheson & Grants' Handbook for Engineers. Second edition, royal 32mo, sewed, gilt edges	2	0
Merrett, H. S. Practical Treatise on the Science of Land and Engineering	12	6

Surveying. Fifth edition, revised by G.
W. USILL. Royal 8vo

Middleton, R. E., & Chadwick, O. A
Treatise on Surveying. Two vols., royal
8vo, *each*

10 6

Molesworth, Sir G. L., & H. B. A
Pocket Book of Useful Formulæ and
Memoranda for Civil and Mechanical
Engineers. Twenty-fourth edition,
32mo, roan, gilt edges

6 0

Molesworth & Hurst. The Pocket Books
of SIR G. L. MOLESWORTH & J. T. HURST,
printed on India paper and bound in one
Vol. Royal 32mo, Russia, gilt edges

12 6

Newman, J. Metallic Structures; Corrosion
and Fouling and their Prevention.
Crown 8vo

9 0

— Earthwork Slips and Subsidences on
Public Works. Crown 8vo

7 6

— Scamping Tricks and Odd Knowledge
occasionally practised upon Public
Works. Crown 8vo

2 6

Penman, W. Land Surveying on the
Meridian and Perpendicular System.
8vo

8 6

Roberts, A. H. Diagrams for the Graphic
Calculation of Earthwork Quantities.
Ten cards, fcap., in cloth case *net*

10 6

Spons' Dictionary of Engineering, Civil,
Mechanical, Military & Naval. Eight
divisions, super-royal 8vo

5 8 0

Complete in 4 Vols. with supplement	7	5	0
Ditto ditto half Mor., top edge gilt	9	0	0
Stanley, W. F. Surveying and Levelling Instruments, theoretically and practically described. Third edition, crown 8vo	7	6	
Tate, J. S. Surcharged and different Forms of Retaining Walls. 8vo, sewed	2	0	
Tudsberry, J. H. S. & Brightmore, A. W. The Principles of Waterworks Engineering. Second edition, medium 8vo	1	5	0
Wells, W. H. Hints on Levelling Operations. Second edition, 8vo sewed	1	0	
Wheeler, W. H. The Drainage of Fens and Low Lands by Gravitation and Steam Power. 8vo	12	6	
Willcocks, W. Egyptian Irrigation. Second edition, royal 8vo	1	10	0

DRAWING.

GEOMETRY—GRAPHIC STATICS.

Alexander, B. The Ornamental Penman's, Engraver's, Sign Writer's, and Stonecutter's Pocket Book of Alphabets. Oblong 12mo, sewed	6		
Andre, G. G. The Draughtsman's Handbook of Plan and Map Drawing. Crown 4to	9	0	

Atkinson, E. H. de V. A Text Book of Practical Solid Geometry. 8vo	7	6
Binns, W. Elementary Treatise on Orthographic Projection. Twelfth edition, 8vo	9	0
—— Second Course of Orthographic Projection. Fourth edition, 8vo	10	6
Clarke, G. S. The principles of Graphic Statics Third edition, 4to	12	6
Cromwell, J. H. A System of Easy Lettering. Fourth thousand, oblong 8vo	2	6
Halliday, G. A First Course of Mechanical Drawing (Tracing), oblong 4to, sewed	2	0
Ott, K. Von. The Elements of Graphic Statics translated by G. S. CLARKE. Crown 8vo	5	0
Stanley, W. F. A Descriptive Treatise on Mathematical Drawing Instruments. Seventh edition, crown 8vo	5	0

ELECTRICAL ENGINEERING.

Allsop, F. C. Electric Bell Construction. Second edition, crown 8vo	3	6
—— Induction Coils and Coil Making. Third edition, crown 8vo	3	6
—— Practical Electric Bell Fitting. Ninth edition. Crown 8vo	3	6
—— Telephones: their Construction and Fitting. Sixth edition, crown 8vo	3	6

Andrews, T. Thermo-electric Reactions and Currents between Metals in Fused Salts. 8vo, sewed	1	0
Crocker, F. B. Electric Lighting: A Practical Exposition of the Art.		
Vol. I. The Generating Plant. Royal 8vo	12	6
Vol. II. Distributing Systems & Lamps. Royal 8vo	12	6
Cushing, Jr., H. C. Standard Wiring for Electric Light and Power. New edition, crown 8vo, limp leather <i>net</i>	4	6
Fahie, J. J. A History of Electric Telegraphy to the year 1837. Crown 8vo	9	0
Fleming, J. A. Magnets and Electric Currents. Crown 8vo	7	6
Halliday, Geo. Notes on design of Small Dynamo. Second edition, 8vo	2	6
Haskins, C. D. Transformers: their Theory, Construction, and Application simplified. Crown 8vo	4	6
Heaphy, M. The Phoenix Fire Office Rules for Electric Light Installation and Electrical Power Installation. Twenty- sixth edition, 8vo, sewed		6
Hedges, K. American Electric Street Railways: their Construction and Equipment. Post 4to	12	6
Hoskiør, Col. V. Testing Telegraph Cables. Third edition, crown 8vo	4	6

Hughes, N. The Magneto-Hand Telephone. Square 16mo	3	6
Inst. of Electrical Engrs. General Rules recommended for Wiring for the Supply of Electrical Energy. 8vo, sewed		6
Kempe, H. R. A Handbook of Electrical Testing. Sixth edition, demy 8vo	18	0
Langdon, W. E. The Application of Electricity to Railway Working. 8vo	10	6
Lintern, W. The Motor Engineer's & Electrical Worker's Handy-book. 12mo <i>net</i>	2	0
Maycock, W. P. Practical Electrical Notes and Definitions. Second edition, royal 32mo	2	0
	3	0
Nichols, E. L. The Galvanometer. Royal 8vo, sewed	4	6
Norrie, H. T. Induction Coils. Second edition, crown 8vo <i>net</i>	4	6
Parker, H. C. A Systematic Treatise on Electrical Measurements. 8vo	4	6
Practical Electrics. A Universal Handybook on Every Day Electrical Matters. Sixth edition, 8vo	3	6
Reagan, H. C. Electrical Engineers' & Students' Chart and Handbook of the Brush Arc Light System. 8vo	4	6

Reed's Electric Lighting for Steamers and its Management. Crown 8vo <i>net</i>	2	6	
Sprague, J. T. Electricity: its Theory, Sources and Applications. Third edition, crown 8vo	15	0	
Thom, C., & Jones, W. H. Telegraphic Connections. Oblong 8vo	7	6	
Thompson, E. P., & Anthony, W. A. Röntgen Rays and Phenomena of the Anode and Cathode. 8vo	7	6	
Thompson, S. P. Dynamo Electric Machinery. <i>Seventh Edition in preparation</i>			
—— Latest Dynamo Electric Machines. Demy 8vo	4	6	
—— The Electro-Magnet and Electro Magnetic Mechanism. <i>Third Edition in the Press</i>			
—— PHILIPP REIS, Inventor of the Telephone: a Biographical Sketch. 8vo, cloth	7	6	
—— Polyphase Electric Currents and Alternate Current Motors. Second edition, demy 8vo	1	1	0
Thompson, S. P., & Thomas, E. Electrical Tables and Memoranda. 64mo, roan, gilt edges	1	0	
Ditto ditto, in celluloid case	1	6	

HOROLOGY

Britten, F. J. The Watch and Clock Maker's Handbook, Dictionary and Guide. Ninth Edition. Crown 8vo <i>net</i>	5	0
—— Former Watch and Clock Makers and their Work. Crown 8vo <i>net</i>	5	0
—— Old Clocks & Watches & their Makers. Demy 8vo <i>net</i>	10	0
—— The Springing & Adjusting of Watches. Crown 8vo <i>net</i>	3	0
Immisch, M. Prize Essay On the Balance Spring and its Isochronal Adjustments. Crown 8vo	2	6
Nelthropp, H. L. Treatise on Watch work, Past and Present. Crown 8vo	6	6

HYDRAULICS.

PUMPS, WATER WHEELS.

Bjorling, P. R. Pumps: Historically, Theoretically and Practically Considered. Second edition, crown 8vo	7	6
Bjorling, P. R. Pump Details. Crown 8vo	7	6
—— Practical Handbook on Direct-Acting Pumping Engine and Steam Pump Construction. Crown 8vo	5	0
—— Pumps and Pump Motors: A Manual for the use of Hydraulic Engineers. Two vol. royal 4to	3	3 0
—— Practical Handbook on Pump Construction. Second edition, crown	5	0

8vo			
—— Water or Hydraulic Motors. Crown 8vo	9	0	
Blaine, R. G. Hydraulic Machinery, with an Introduction to Hydraulics. Demy 8vo	14	0	
Box, T. Practical Hydraulics. Twelfth edition, crown 8vo	5	0	
Colyer, F. Hydraulic, Steam, and Hand Power Lifting and Pressing Machinery. Second edition, imperial 8vo	1	8	0
—— Pumps and Pumping Machinery——			
Vol I. Second edition, 8vo	1	8	0
Vol II. Second edition, 8vo	1	5	0
Cullen, W. Practical Treatise on the Construction of Horizontal and Vertical Waterwheels. Second edition, small 4to	5	0	
Davey, H. Description of the Differential Expansive Pumping Engine. 8vo	2	0	
De Salis, R. Tables giving Hydraulic Mean Depth & Area of Circular Sewers. 8vo, sewed	1	0	
Donaldson, W. Donaldson's Poncelet Turbine and Water Pressure Engine and Pump. 4to	5	0	
—— Principles of Construction and Efficiency of Waterwheels. 8vo	5	0	
Hennell, T. Hydraulic and other Tables for purposes of Sewage & Water Supply. Crown 8vo	5	0	

Hick, J. Experiments on the Friction of the Leather Collar in Hydraulic Presses. 8vo, sewed	1	0
Higham, T. Hydraulic Tables for Finding the Mean Velocity and Discharge in Open Channels. Second edition, super- royal, 8vo	7	6
Silk, A. E. Tables for Calculating the Discharge of Water in Pipes for Water & Power Supplies. Indexed at side for ready reference. Crown 8vo	5	0
Stone, T. W. Simple Hydraulic Formulæ. Crown 8vo	4	0

INDUSTRIAL CHEMISTRY AND MANUFACTURES.

Bater, C. H. Brewing Calculations, Gauging & Tabulation. 64mo, roan, gilt edges	1	6
Bayley, T. A Pocket Book for Chemists, Chemical Manufacturers, Metallurgists, Dyers, Distillers, etc. Seventh edition, Royal 32mo, roan, gilt edges	5	0
—— A Pocket Book for Pharmacists, Medical Practitioners, Students, etc. Royal 32mo, roan, gilt edges	5	0
Brooks, C. P. Cotton Manufacturing. Crown 8vo, cloth <i>net</i>	2	6
Brooks, C. P. Weaving Calculations. Crown 8vo, cloth	2	6

Burns, W. Illuminating and Heating Gas. A Manual of Manufacture of Gas from Tar, Oil etc., etc. Crown 8vo	5	0
Carpenter, W. L., & Leask, H. A Treatise on the Manufacture of Soap and Candles, Lubricants and Glycerine. Second edition, crown 8vo	12	6
Cross, C. F., & Bevan, E. J. A Text Book of Paper Making. Second edition, crown 8vo	12	6
Deerr, N. Sugar House Notes and Tables. 8vo <i>net</i>	10	6
Dubelle, G. H. The “non plus ultra” Soda Fountain Requisites of Modern Times. A Practical Receipt Book for Druggists, Chemists, etc. Crown 8vo <i>net</i>	4	6
Foster, J. Treatise on the Evaporation of Saccharine, Chemical & other Liquids by the Multiple System in Vacuum and Open Air. Second edition, 8vo	1	1 0
Graham, D. A. A Treatise on the Comparative Commercial Values of Gas Coals and Cannels. 8vo	4	6
Graham, M. Practical Hints on the Working and Construction of Regenerator Furnaces. fcap. 8vo., leather	3	0
Hansen, E. C. Practical Studies in Fermentation, translated from the German by A. K. Miller. Ph. D. 8vo	12	6

Hanssen, C. J. T. Reform in Chemical and Physical Calculations. Demy 4to <i>net</i>	6	6	
Hodgetts, E. A. B. Liquid Fuel for Mechanical and Industrial Purposes. 8vo	5	0	
Hornby, J. The Gas Engineer's Laboratory Handbook. Crown 8vo	6	0	
Ingle, H. & H. The Chemistry of Fire and Fire Prevention. Crown 8vo	9	0	
Lee, D. Manual for Gas Engineering Students. 18mo	1	0	
Lovibond, T. W. Brewing with Raw Grain. Crown 8vo	5	0	
Procter, H. R. Leather Industries: Laboratory Book on Analytical & Experimental Methods. Demy 8vo	9	0	
Redwood, I. I. Theoretical and Practical Ammonia Refrigeration. Third edition, square 16mo	4	6	
Richards, W. Practical Treatise on the Manufacture and Distribution of Coal Gas. Demy 4to	1	8	0
Scammell, G., & Collyer, F. Breweries and Maltings. Second edition, 8vo	12	6	
Speyers, C. L. Text Book of Organic Chemistry. Demy 8vo	9	0	
Spons' Encyclopædia of the Industrial Arts, Manufactures and Commercial Products. Super-royal 8vo			

In 2 Vols., cloth	3	10	0
In 5 Divisions, cloth	3	11	6
In 2 Vols. half Morocco, top edge gilt	4	10	0
Standage, H. C. The Practical Polish and Varnish Maker. Crown 8vo		6	0
Streatfield, F. W. Practical Work in Organic Chemistry. Crown 8vo		3	0
Terry, G. Pigments, Paints and Painting. Crown 8vo		7	6
Van Eijndhoven, A. J. A Comparison of the English and French Methods of Ascertaining the Illuminating Power of Coal Gas. Crown 8vo		4	0
Wein, E., & Frew, W. Tables for the Quantitative Estimation of the Sugars. Crown 8vo		6	0
Workshop Receipts. For the use of Manufacturers, Mechanics and Scientific Amateurs. Crown 8vo, 5 Vols each		5	0
First Series. Miscellaneous.			
Second Series. Chemical Manufactures.			
Third " Electrical & Metallurgical.			
Fourth " Handicrafts and Mechanical Subjects.			
Fifth " Miscellaneous.			

MARINE ENGINEERING AND NAVAL ARCHITECTURE.

Barnaby, S. W. Marine Propellers. Fourth edition, demy 8vo <i>net</i>	10	6
Bartley, B. C. The Marine Engineer's Record Book: Engines. 8vo, roan <i>net</i>	5	0
The Engineer's and Draughtsman's Data Book for Workshop and Office use. Third edition, crown 8vo, roan	3	0
Franklin, A. C. Yachting Hints, Tables and Memoranda. Waistcoat pocket size, 64mo, roan, gilt edges	1	0
Ditto ditto in celluloid case	1	6
Haldane, J. W. C. Steamships and their Machinery from first to last. 8vo	15	0
Hogg, A. Tables for Constructing Ships' Lines. Second edition, 8vo	7	0
Hovgaard, G. W. Submarine Boats. Crown 8vo	5	0
Jordan, C. H. Tabulated Weights of Angle, Tee, Bulb, Round, Square, and Flat Iron and Steel for the use of Naval Architects, Ship-builders, &c. Fifth edition, French morocco, gilt edges	7	6
Jordan, C. H. Particulars of Dry Docks on the river Thames. On a sheet folded in cloth. 18mo	2	6
Little, H. The Marine Transport of Petroleum. Crown 8vo	10	6
Main, J. The Progress of Marine Engineering from the time of Watt to the present day. Crown 8vo	7	6

Reed's Engineers' Handbook to the Board of Trade Examinations for certificates of Competency as First and Second Class Engineers. Seventeenth edition, 8vo <i>net</i>	14	0
Reed. The key to the Seventeenth edition of REED'S Engineers' Handbook, 8vo <i>net</i>	7	6
—— Examination Papers for Extra First Class Engineers. Third edition, 8vo <i>net</i>	14	0
—— Polyglot Guide to the Marine Engine in English, French, German and Norsk. Oblong 8vo <i>net</i>	10	0
—— Marine Boilers. Second edition, crown 8vo <i>net</i>	4	6
—— Useful Hints to sea-going Engineers. Third edition, crown 8vo <i>net</i>	3	6
—— Drawing of the Triple Expansion Engine and the Pipe Arrangements. On 2 sheets in case <i>net</i>	3	6
—— Drawing of a Compound Engine in case <i>net</i>	1	6

MATERIALS.

Box, T. Practical Treatise on the Strength of Materials. Third edition, crown 8vo	18	0
Britton, T. A. Treatise on the Origin, Progress, Prevention and Cure of Dry Rot in Timber. Crown 8vo	7	6
Butler, D. B. Portland Cement: its Manufacture, Testing and Use. 8vo	18	0

Calcare. Cement Users' and Buyers' Guide. 32mo, leather <i>net</i>	2	6
Delano, W. H. Twenty years' practical experience of Natural Asphalt and Mineral Bitumen. Crown 8vo, parchment	2	0
Grant, J. Experiments on the Strength of Cement. 8vo	10	6
Heath, A. H. A Manual of Lime and Cement. Crown 8vo	6	0
Luard, C. E. Stone: how to get it and how to use it. 8vo	2	0
Newman, J. Notes on Concrete and Works in Concrete. Second edition, crown 8vo	6	0
Paterson, M. M. The Testing of Pipes and Pipe-joints in the open trenches. 8vo, sewed	2	0
Redwood, I. I. Lubricants. Oils and Greases. 8vo	4	6
——— A Practical Treatise on Mineral Oils and their By-Products. Demy 8vo	15	0
Stoffler, E. Silico-Calcareous Sandstones. 8vo, sewed <i>net</i>	4	0

MATHEMATICS.

Blaine, R. G. Quick and Easy Methods of Calculating. 16mo, leather	2	6
--	---	---

Byrne, O. General Method of Solving Equations of all Degrees. 8vo, sewed	1	0
Graham, J. Elementary Treatise on the Calculus for Engineering Students. Second edition, crown 8vo	7	6
Higgs, P. Algebra, Self-Taught. Third edition, crown 8vo	2	6
Ormsby, M. T. Elementary Practical Mathematics. Demy 8vo <i>net</i>	7	6
Sloane, T. O'c. The Arithmetic of Electricity. Crown 8vo	4	6

MECHANICAL ENGINEERING.

STEAM ENGINES, STEAM BOILERS, GAS ENGINES, ETC.

Adams, Hy. Handbook for Mechanical Engineers. Fourth edition, crown 8vo	7	6
Applebys' Handbooks of Machinery. In six sections, 8vo, each	3	6
Section 1.—Prime Movers, Steam, Gas and Air Engines, Boilers, Turbines, &c.		
Section 2.—Hoisting Machinery, Winding Engines and Lifting Appliances.		
Section 3.—Pumping Machinery, Pumping Engines, Centrifugal, Steam, Electrical and Hand Pumps.		
Section 4.—Machine Tools and Accessories.		
Section 5.—Contractors' Plant and Railway Materials. (<i>In the Press.</i>)		
Section 6.—Mining, Colonial and Manufacturing Machinery.		
The portion of Section 6, dealing with Mining Machinery, is to be had separately, price 2s. 6d.		

Barber, T. W. Engineers' Sketch Book of Mechanical Movements. Third edition, 8vo	10	6
—— The Repair & Maintenance of Machinery. 8vo	10	6
Box, T. Practical Treatise on Mill Gearing. Fifth edition, crown 8vo	7	6
Byrne, O. The Essential Elements of Practical Mechanics. Fourth edition, post 8vo	7	6
Campin, F. The Practice of Hand Turning. Third edition, crown 8vo	3	6
Colyer, F. The Working and Management of Steam Boilers and Engines. Second edition, crown 8vo	3	6
—— Treatise on Modern Steam Engines and Boilers. 4to	12	6
Cooper, J. H. Treatise on the use of Belting for the Transmission of Power. Fourth edition, demy 8vo	15	0
Cotterill, J. H. The Steam Engine considered as a Thermodynamic Machine. Third edition 8vo	15	0
Dahlstrom, K. P. The Fireman's Guide, a Handbook on the Care of Boilers. Eighth edition, fcap. 8vo	2	0
Diesel, R. Theory & Construction of a Rational Heat Motor. 8vo	6	0
Donaldson, W. Transmission of Power by Fluid Pressure, Air and Water. 8vo	6	0

Fletcher, W. History and Development of Steam Locomotion on Common Roads. 8vo	5	0
Foden, J. The Boiler Makers' and Iron Ship-builders' Companion. Fourth edition, fcap. 8vo	5	0
Fuller, J. Art of Copper Smithing. Royal 8vo	12	6
Goldingham, A. H. The Design & Construction of Oil Engines. Crown 8vo <i>net</i>	6	0
Graham, J. C. Elementary Treatise on Steam and the Use of the Indicator. 8vo	9	0
Halliday, G. Belt Driving. 8vo	3	6
Hanssen, A. The Commercial Efficiency of Steam Boilers. Large 8vo, sewed		6
Henthorn, J. T. The Corliss Engine. Third edition, square 16mo	3	6
Hett, C. L. Table of the Power of Leather Belting and Shafts. On folding card	1	0
Hiscox, J. D. Gas, Gasoline and Oil Vapour Engines. Fourth edition, 8vo <i>net</i>	10	6
Kinealy, J. H. Elementary Text-book on Steam Engines and Boilers. Third edition, 8vo	10	6
Knight, C. The Mechanician: a Treatise on the Construction and Manipulation of Tools. Fifth edition, 4to	18	0

Lieckfeld, G. Practical Handbook on the Care and Management of Gas Engines. Square 16mo	3	6	
Low, P. A. Valve Setting Record Book. 8vo boards	1	6	
Millis, C. T. Metal Plate Work, its Patterns and their Geometry. Third edition, crown 8vo	9	0	
Peattie, J. Steam Boilers, their Management and Working. Fourth edition, crown 8vo	5	0	
Porter, C. T. Treatise on the Richards Steam Engine Indicator. Fifth edition, 8vo	9	0	
Richards, J. The Arrangement, Care and Operation of Wood Working Factories and Machinery. Second edition, crown 8vo	5	0	
Rigg, A. Practical Treatise on the Steam Engine. Second edition, demy 4to	1	5	0
Sexton, M. J. Pocket Book for Boiler Makers and Steam Users. Fourth edition, royal 32mo, roan, gilt edges	5	0	
Spencer, A. Roll Turning for Sections in Steel and Iron. Second edition, 4to	1	10	0
Uhland, W. H. Slide and Piston-Valve Geared Steam Engines. Two Vols., half Morocco	1	16	0
Watson, E. P. How to run Engines and Boilers. Fourth edition	3	6	

Welch, E. J. Practical method of Designing Slide Valve Gearing. Crown 8vo		6	0
Wright, T. W. Elements of Mechanics. 8vo	1	10	6
Zeuner, G. Treatise on Valve Gears, translated from the fourth German edition by J. F. KLEIN. 8vo		12	6

METALLURGY.

Andrews, T. The Life of Railway Axles. 8vo, sewed		1	0
—— Microscopic Internal Flaws in Steel Rails and Propellor Shafts. 8vo, sewed		1	0
—— Microscopic Internal Flaws, Inducing Fracture in Steel. 8vo, sewed		2	0
Davies, J. Galvanized Iron: its Manufacture and Uses. 8vo <i>net</i>		5	0
Ede, G. Guns and Gun Making Material. Crown 8vo		6	0
—— The Management of Steel. Seventh edition, crown 8vo		5	0
Kirk, E. The Cupola Furnace. Demy 8vo		14	0
Macfarlane, J. W. Practical Notes on Pipe Founding. 8vo		12	6
Sharp, J. Modern Foundry Practice. 8vo <i>net</i>	1	1	0
West, T. D. The Metallurgy of Cast Iron. Second edition. 8vo		12	6

Wyllie, C. Treatise on Iron & Steel Founding. Second edition, crown 8vo	5	0
---	---	---

MINERALOGY AND MINING.

Andre, G. G. Rock Blasting, 8vo	5	0
--	---	---

—— Practical Treatise on Coal Mining. Two Vols., royal 4to	3	12 0
---	---	------

Brown, W. L. Manual of Assaying Gold, Silver, Copper and Lead Ores. Ninth edition, crown 8vo	12	6
---	----	---

Charleton, A. G. Tin: Describing the Chief Methods of Mining, Dressing, etc. Crown 8vo	12	6
---	----	---

Daw, A. W., & Z. W. The Blasting of Rock in Mines, Quarries and Tunnels, etc. Demy 8vo	15	0
---	----	---

Hopton, W. Conversations on Mines. Ninth edition, crown 8vo	4	6
---	---	---

Hull, E. Our Coal Resources at the End of the Nineteenth Century. Demy 8vo	6	0
--	---	---

Kirkpatrick, T. S. G. Simple Rules for the Discrimination of Gems	2	0
---	---	---

—— The Hydraulic Gold Miners' Manual. Second edition, crown 8vo	4	0
--	---	---

Lock, C. G. W. Economic Mining. 8vo	1	1 0
--	---	-----

—— Gold Milling: Principles and Practice. Demy 8vo <i>net</i>	1	10 0
--	---	------

—— Mining and Ore-Dressing Machinery. Super-royal 4to	1	5	0
—— Miner's Pocket Book. Fourth edition, fcap. 8vo, roan <i>net</i>	12	6	
Longridge, C. C. The Holloway Longridge Process for Extracting Gold. 8vo, sewed	1	6	
Miller, J. A. The Practical Handbook for the Working Miner and Prospector, and the Mining Investor. Crown 8vo	7	6	
Murgue, D. The Theory and Practice of Centrifugal Ventilation Machines. 8vo	5	0	
Povey-Harper, J. Examples of Coal Milling Plant. Second edition, in portfolio	4	4	0
Suttie, T. R. The Miner's and Prospector's Pocket Guide. Square 16mo <i>net</i>	2	0	

ORGANIZATION AND MANAGEMENT.

Brown, N. The Organization of Gold Mining Business. Fcap. folio <i>net</i>	1	5	0
Foster, H. A. Central Station Management and Finance. 8vo	7	6	
—— Central Station Book Keeping. 8vo	12	6	
Lewis, J. S. The Commercial Organization of Factories. Royal 8vo. <i>New Edition in the Press.</i>			
Matheson, E. Depreciation of Factories. 2nd edition, 8vo	7	6	

—— Aid Book to Engineering Enterprise. Third edition, 8vo	1	4	0
Parry, W. Kaye. Office Management. A handbook for Architects and Civil Engineers. <i>net</i>	10	0	
Williams, R. R. The American Hardware Store. Royal 8vo	12	6	

RAILWAY ENGINEERING.

Allen, C. F. Railroad Curves and Earthwork. 12mo, leather, gilt edges <i>net</i>	8	6	
Allen, G. T. Tables of Parabolic Curves for the use of Railway Engineers and others. Fcap. 16mo	4	0	
Blackall, R. H. Up to date Air Brake Catechism. Crown 8vo <i>net</i>	6	0	
Briggs, C. Simple and Automatic Vacuum Brakes. 8vo	4	0	
Cole, W. H. Notes on Permanent-way Material, Plate-laying, and Points and Crossings. Third edition, crown 8vo	7	6	
Derr, W. L. Block Signal Operation. Oblong crown 8vo	7	6	
Glover, J. Formulæ for Railway Crossings and Switches. Royal 32mo	2	6	
Grimshaw, R. Locomotive Catechism. Crown 8vo	7	6	

Grover, J. W. Estimates and Diagrams of Railway Bridges. Second edition, folio	1	11	6
—— Iron and Timber Railway Superstructures and General Works. Folio	2	2	0
—— Examples of Station Buildings and their Cost. Folio		10	6
Haldane, J. W. C. Railway Engineering, Mechanical and Electrical. Demy 8vo		15	0
Hogg, C. P. Tables for setting-out Railway Curves. A series of cards in neat cloth case		4	6
Hughes, G. The Construction of the Modern Locomotive. 8vo		9	0
Johnson, J. R. Practical Hints for Light Railways at Home and Abroad. Crown 8vo		2	6
Jones, T. W. Permanent Way Pocket Book. Oblong 8vo		4	6
Kennedy, A., & Hackwood, R. W. Tables for setting out Curves for Railways, Roads, Canals, etc. 32mo		2	6
Macgregor, W. Tables for Computing the Contents of Earthwork in the Cuttings and Embankments of Railways. Royal 8vo		6	0
Paterson, J. Tables and Diagrams of Switches and Crossings. 8vo		3	6
Rapier, R. C. Remunerative Railways for New Countries. Crown 4to		15	0

Spooner, C. E. Narrow Gauge Railways. Second edition, 8vo	15	0
---	----	---

Watson, A. G. Practical Hints on setting out Curves. 18mo	6	0
---	---	---

SANITATION.

Bailey-Denton, E. Sewage Purification. 8vo	5	0
--	---	---

Birch, R. W. P. Sewage Irrigation by Farmers. 8vo, sewed	2	6
--	---	---

Boulnois, H. P. Dirty Dustbins and Sloppy Streets. Crown 8vo, sewed	1	0
---	---	---

—— The Municipal and Sanitary Engineer's Handbook. Third edition, demy 8vo	15	0
---	----	---

Coleman, T. E. Sanitary House Drainage, its Principles and Practice. Crown 8vo	6	0
--	---	---

—— Stable Sanitation & Construction. Crown 8vo	6	0
---	---	---

Colyer, F. Public Institutions, their Engineering, Sanitary, and other Appliances. 8vo	5	0
---	---	---

—— Treatise on the Modern Sanitary Appliances for Healthy Residences. Crown 8vo	5	0
---	---	---

Davies, J. P. Standard Practical Plumbing.		
Vol I. Fourth edition, royal 8vo	7	6
Vol II. Royal 8vo <i>net</i>	10	6

Davis, G. B., & Dye, F. A Complete and Practical Treatise on Plumbing & Sanitation. 2 vols., 4to cloth <i>net</i>	2	15	0
Parsons, Hon. R. C. Las Obras de Salubridad de la Ciudad de Buenos Aires. 8vo <i>net</i>		5	0
Robinson, H. Sewerage and Sewage Disposal. Second edition, demy 8vo		5	0
Smeaton, J. Plumbing, Drainage, Water Supply and Hot Water Fitting. 8vo		7	6

WARMING AND VENTILATION.

Box, T. Practical Treatise on Heat. Ninth edition, crown 8vo	12	6	
Drysdale, J., & Hayward, J. W. Health and Comfort in House Building, or Ventilation with warm air by self-acting suction power. Third edition, 8vo	7	6	
Dye, F. Hot Water & Steam Cooking Apparatus. Second edition, 18mo	1	6	
—— Hot Water Supply. Fourth edition, crown 8vo	3	0	
—— A Practical Treatise upon Steam Heating. Demy 8vo <i>net</i>	10	0	
Eldridge, J. The Engineer's Practical Guide for Fixing Hot Water Apparatus. Second edition, crown 8vo, sewed	1	0	
Hood, C. Practical Treatise upon Warming Buildings by Hot Water. Third edition,	15	0	

revised by F. DYE. Demy 8vo

Kinealy, J. H. Charts for Low Pressure Steam Heating. Small folio	4	6
—— Formulæ and Tables for Heating. 8vo	3	6
Rafter, G. W. Mechanics of Ventilation. 2nd edition, 18mo, boards	2	0

WATER SUPPLY.

Collet, H. Water Softening and Purification. Crown 8vo	5	0
Colyer, F. Treatise on Water Supply, Drainage & Sanitary Appliances of Residences. Crown 8vo	5	0
Hill, J. W. Purification of Public Water Supplies. 8vo	10	6
Parry, J. Water Supply. New edition. Crown 8vo	3	6
Richert, G. On Artificial Underground Water. 8vo, sewed <i>net</i>	1	0
Spon, E. Well Sinking. <i>New Edition in the Press</i>		
Stone, F. W. Notes on Water Supply. Crown 8vo	5	0
Tudsbery, J. H. T., & Brightmore, A. W. The Principles of Waterworks' Engineering. Second edition, 8vo	1	5 0

USEFUL TABLES.

Babbage, C. Tables of Logarithms of the Natural Numbers from 1 to 108,000. Stereotype edition. 8vo	1	7	6
Barlow's Tables of Squares, Cubes, Square Roots, Cube Roots & Reciprocals. Crown 8vo	1	6	0
Brook, J. French Measures & English Equivalents. Fcap. 32mo, roan	1	1	0
Clark, L. A Dictionary of Metric and other useful measures. 8vo	1	6	0
Girder, W. J. Tables of the Weight of Iron. On large folding card	1	1	0
Keerayeff, P. Tables of some of the Principal Speeds occurring in Mechanical Engineering, expressed in Metres per second. 18mo, sewed	1	6	
La Nicca, J. Turner's and Fitter's Pocket Book. 18mo, sewed	1	6	
Lindsay, Lord. Tables for Engineers and Mechanics, giving the values of the different trains of wheels required to produce Screws of any pitch. Second edition, royal 8vo, oblong	1	2	0
Martin, W. A. Screw-cutting Tables. Seventh edition, royal 8vo, oblong	1	1	0
Molesworth, Sir G. L. Metrical Tables. Third edition, royal 32mo	1	2	0
Price, W. Turner's Handbook on Screw- cutting, Coning, etc. Fcap. 8vo	1	1	0

Reed's Screw and Worm-wheel Cutting up to date. Royal 32mo	1	2	6
Rownson's Iron Merchant's Tables and Memoranda, Weights and Measures. 32mo, leather	1	3	6
Spons' Tables and Memoranda for Engineers, by J. T. HURST, C. E. Twelfth edition, 64mo, roan, gilt edges	1	1	0
Ditto ditto, in celluloid case	1	1	6
Taylor, T. Gauges at a Glance. Second edition. Post 8vo, oblong, with tape converter <i>net</i>	1	5	0
Thompson, S. P. Optical Tables and Data, for the use of Opticians. Oblong 8vo <i>net</i>	1	6	0
Unwin, W. C. Short Logarithmic and other Tables. Fourth edition, small 4to	1	3	0
Whitelaw, H. N. Fifty-four hours' Wages Calculator. Second edition, 8vo <i>net</i>	1	2	6
Woodward, C. J. A. B. C. Logarithms for general use, with lateral index for ready reference. Crown 8vo, limp leather	1	4	0

MISCELLANEOUS.

Boulnois, H. P. Practical Hints on Taking a House. 18mo	1	6	
Boulvin, M. J. The Entropy Diagram and its Applications. Demy 8vo	5	0	

Burr, S. D. V. Bicycle Repairing. Fourth edition, 8vo	4	6
Dye, F. The Cooking Range: its Failings and Remedies. Fcap. 8vo, sewed		6
Gillett, W. The Phonograph, and how to construct it. Crown 8vo	5	0
Hopkins, G. M. Experimental Science, Elementary, Practical and Experimental Physics. 20th edition, large 8vo	16	0
Longridge, J. A. Internal Ballistics. 8vo	18	0
——— Treatise on the Application of Wire to the Construction of Ordnance. 8vo	1	5 0
——— The Progress of Artillery: Naval Guns. 8vo, sewed	2	0
——— The Field Gun of the Future. 8vo, sewed	2	6
Philipson, W. Prize Essay on the Suspension of Carriages. 8vo	5	0
Richards, W. The Gas Consumer's Handy Book. 18mo, sewed		6
Spons' Engineer's Diary and Year Book, issued Annually. 4to	3	6
——— Mechanic's Own Book: A Manual for Handicraftsmen and Amateurs. Fifth edition, demy 8vo	6	0
Ditto ditto half Morocco	7	6
Vernon, A. Estate Fences: their Choice, Construction and Cost. 8vo	15	0

A SUPPLEMENT
TO
SPONS' DICTIONARY OF ENGINEERING,
CIVIL, MECHANICAL, MILITARY, AND NAVAL.

EDITED BY ERNEST SPON,
ASSOC. MEM. INST. C.E., MEM. SOC. ENGINEERS, OF THE FRANKLIN INSTITUTE, AND
OF THE GEOLOGISTS' ASSOCIATION.

In 18 Parts, price 2s. each, post free 2s. 2d.
Bound in cloth, 3 Divisions, 13s. 6d. each.
Or, in One Vol., cloth, £2; half-morocco, £2 8s.

COMPLETE LIST OF ALL THE SUBJECTS.

	Part
Abacus	1
Agricultural Implements	1
Air Compressors	1, 2
Animal Charcoal Machinery	2
Antimony	2
Axles & Axle-boxes	2
Barn Machinery	2
Belts and Belting	2
Blasting	3
Boilers	3
Brake	3

Brick-making Machine	3, 4
Bridges	4, 5
Cages	5
Calculus	5
Canals	5
Carpentry	5
Cast Iron	5, 6
Cement, Concrete, Limes & Mortar	6
Chimney Shafts	6
Coal Cleansing and Washing	6
Coal Mining	6, 7
Coke Ovens	7
Copper	7
Dock	7, 8
Drainage	8
Dredging	8
Dynamo-Electric and Magneto-Electric Machines	8
Dynamometer	8, 9
Electrical Engineering	9, 10
Engines, Varieties of	10
Explosives	10
Fans	10
Founding	10, 11
Gas, Manufacture of	11
Hammers	11
Heat	12
Horse Power	12
Hydraulics	12
Hydro-geology	12
Indicator	12, 13
Iron	13
Lifts, Hoists and Elevators	13
Lights, Buoys and Beacons	13, 14
Machine Tools	14

Materials of Construction, Strength of	14, 15
Mercury	15
Meters	15
Ores	15
Piers	15
Pile Driving	15
Pneumatic Transmission	15
Pump	15
Pyrometer	15
Road Locomotive	15, 16
Rock Drill	16
Rolling Stock	16, 17
Sanitary Engineering	17, 18
Shafts and Shaft Fittings	18
Steel	18
Stone-working Machinery	18
Tramways	18
Well Sinking and Boring	18

SPONS' ENCYCLOPÆDIA

OF THE

Industrial Arts, Manufactures and Commercial Products.

EDITED BY C. G. WARNFORD LOCK, F.L.S., &c.

In super-royal 8vo, containing 2100 pages, and Illustrated by nearly 1500 Engravings.

Can be had in the following bindings:

Vols., cloth	£3	10	0
Divisions, cloth	3	11	6
Vols., half-morocco, top edge gilt, bound in a superior manner	4	10	0
} Parts, Sewed, at 2s. each.			

Any Part can be had separate, price 2s.; postage 2d.

COMPLETE LIST OF ALL THE SUBJECTS.

	Part
Acids	1, 2, 3
Alcohol	3, 4
Alkalies	4, 5
Alloys	5, 6
Arsenic	6
Asphalt	6
Aerated Waters	6

Beer and Wine	6, 7
Beverages	7, 8
Bleaching Powder	8
Bleaching	8, 9
Borax	9
Brushes	9
Buttons	9
Camphor	9, 10
Candles	10
Carbon	10
Celluloid	10
Clays	10
Carbolic Acid	11
Coal-tar Products	11
Cocoa	11
Coffee	11, 12
Cork	12
Cotton Manufactures	12, 13
Drugs	13
Dyeing and Calico Printing	13, 14
Dyestuffs	14
Electro-Metallurgy	14
Explosives	14, 15
Feathers	15
Fibrous Substances	15, 16
Floor-cloth	16
Food Preservation	16
Fruit	16, 17
Fur	17
Gas, Coal	17
Gems	17
Glass	17
Graphite	18
Hair Manufacture	18

Hats	18
Ice, Artificial	18
Indiarubber Manufactures	18, 19
Ink	19
Jute Manufactures	19
Knitted Fabrics—Hosiery	19
Lace	19
Leather	19, 20
Linen Manufactures	20
Manures	20
Matches	20, 21
Mordants	21
Narcotics	21, 22
Oils and Fatty Substances	22, 23, 24
Paper	24
Paraffin	24
Pearl and Coral	24
Perfumes	24
Photography	24, 25
Pigments and Paint	25
Pottery	25, 26
Printing and Engraving	26, 27
Resinous and Gummy Substances	26, 27
Rope	27
Salt	27, 28
Silk	28
Skins	28
Soap, Railway Grease and Glycerine	28, 29
Spices	29
Starch	29
Sugar	29, 30, 31
Tannin	31, 32
Tea	32
Timber	32

Varnish	32
Wool and Woollen Manufactures	32, 33

Finsbury Technical Manuals.

EDITOR OF THE SERIES,
Prof. SILVANUS P. THOMPSON,
D.Sc., B.A., F.R.S., M. Inst. E.E., &c.

The Manuals in this series are written by Specialists who are, or have been, on the staff of the City and Guilds Technical College at Finsbury.

Dynamo-Electric Machinery. A Manual for Students of Electro-technics. By Professor SILVANUS P. THOMPSON, D.Sc., F.R.S., &c. (*Seventh Edition in Preparation.*)

Metal Plate Work. Its Patterns and their Geometry. By C. T. MILLIS, M.I.Mech.E. Third edition, considerably enlarged, with 216 illustrations, 406 p.p. 8vo cloth, 9s.

Gas and Petroleum Engines. By Professor WILLIAM ROBINSON, M.E., M.I.E.E., A.M.I.C.E. Illustrated. 8vo, cloth. (*Second Edition in Preparation.*)

Practical Work in Organic Chemistry. By F. W. STREATFIELD, F.I.C., &c. Illustrated. Crown 8vo, cloth, 3s.

The Electromagnet and Electromagnetic Mechanism. By Professor SILVANUS P. THOMPSON, D.Sc., F.R.S., &c. (*Third Edition in Preparation.*)

Polyphase Electric Currents & Alternate Current Motors. By
Professor SILVANUS P. THOMPSON, D.Sc., F.R.S., &c. Second and
enlarged edition. 8vo, cloth, 21s.

An Elementary Treatise on the Calculus for Engineering Students.
By JOHN GRAHAM, B.A., B.E. Second edition, 7s. 6d.

Hydraulic Machinery. With an introduction to Hydraulics. By ROBERT
GORDON BLAINE, Assoc. M. Inst. C.E., &c. With 272 illustrations,
383 p.p. 8vo, cloth, 14s.

The Journal of the Iron and Steel Institute.

Edited by BENNETT H. BROUGH, *Secretary*.

Published Half-yearly, 8vo, cloth, 16s.

The Journal of the Institution of Electrical Engineers.

Edited by W. G. McMILLAN, *Secretary*.

Science Abstracts.

Physics and Electrical Engineering.

Edited by W. R. COOPER, M.A., B.Sc.

Published Monthly, 8vo, 2s. net. Annual Subscription, 24s. net, post free, to any part of the World.

The Transactions of the Society of Engineers.

Edited by PERRY F. NURSEY, *Secretary*.

Published Annually, 8vo, cloth, 15s.

**The Proceedings of the Incorporated Association of Municipal and
County Engineers.**

Edited by THOMAS COLE, ASSOC. M. Inst. C.E.

The Transactions of the Incorporated Institution of Gas Engineers.

Edited by THOMAS COLE, ASSOC. M. Inst. C.E.

Published Annually, 8vo, cloth, 21s.

The Transactions of the Institution of Mining and Metallurgy.

Edited by C. McDERMID, *Secretary*.

Published Annually, 21s. net.

London: E. & F. N. SPON, Ltd., 125, Strand, W.C.

THE PHOSPHOR BRONZE CO. LIMITED,

Sole Makers of the following ALLOYS:

PHOSPHOR BRONZE.

“Cog Wheel” and “Vulcan” Brands. Ingots, Castings, Plates, Strip, Bars and Wire.

“DURO METAL” (*Registered Title*).

For Roll Bearings, Wagon Brasses, &c.

PHOSPHOR TIN AND PHOSPHOR COPPER.

“Cog Wheel” Brand. The best qualities made.

PLASTIC METAL.

“Cog Wheel” Brand. The best filling and lining Metal in the market.

BABBITT’S METAL.

“Vulcan” Brand. Seven Grades.

WHITE LINING METAL, brand “PHOSPHOR,”

Equal to the best White Brass No. 2.

“WHITE ANT” METAL, No. I.

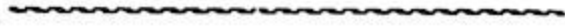
Cheaper than any Babbitt’s, and superior to Magnolia Metal.

“WHITE ANT” BRONZE.

Superior to Fenton’s Metal for Car Bearings.

SILICIUM BRONZE ELECTRICAL WIRE.

For Overhead Electrical Lines and other purposes.



*Please apply for Catalogues containing full particulars to the Company's
Head Office,*

**87 SUMNER STREET, SOUTHWARK,
LONDON, S.E.**

WHITE ANTIFRICTION ALLOYS.

THE BEST, CHEAPEST, AND MOST DURABLE
AND RELIABLE OF THE NUMEROUS COMPOSITIONS
IN THE MARKET IS THE

“WHITE ANT” BRAND
(REGISTERED TRADE MARK)

MADE ONLY BY THE
PHOSPHOR BRONZE Co., LIMITED,
87 SUMNER STREET, SOUTHWARK,
LONDON, S.E.
And at **BIRMINGHAM.**

Also Makers of the following well-known Alloys:

PLASTIC METAL (“Cog Wheel” Brand).

BABBITT’S METAL (“Vulcan” Brand) Seven Grades.

WHITE LINING METAL (Brand “Phosphor.”)

“WHITE ANT” BRONZE

Superior to Fenton’s Metal for Car Bearing.

PHOSPHOR BRONZE (“Cog Wheel” & “Vulcan” Brands.)

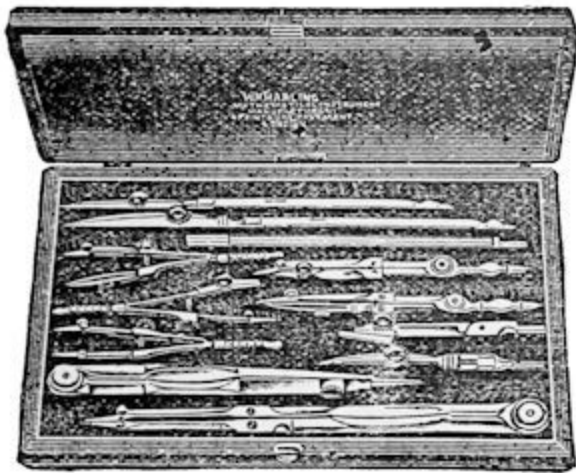
Castings in every kind of Bronze, Brass and Gun Metal, Rolled and Drawn Bronze, Brass, Gun Metal, Tin, German Silver, &c.

W. H. HARLING'S Drawing Instruments

ARE ACKNOWLEDGED BY ALL WHO USE THEM TO BE THE
BEST IN THE MARKET.

VERY GREAT CARE is taken in their manufacture, and EVERY PIECE is examined by a competent workman before leaving the factory.

SPECIAL POCKET CASES, containing good sound instruments of my own manufacture, but not extra finish (recommended):—

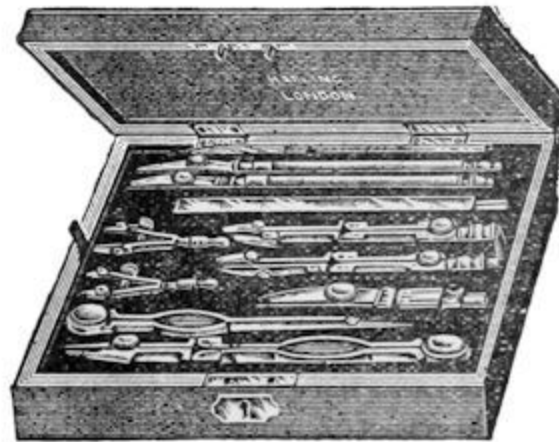


No. 721A.

721A MOROCCO CASE, containing the following electrum instruments:—6-inch double-jointed, needle-pointed compass, ink and pencil points, and lengthening bar—ink and pencil double-jointed, needle-pointed bows—set of three spring bows—hair divider—two drawing pens, and ivory scale, £3 3s.

721B Ditto, ditto, but without set of three spring bows, £2 10s.

717 MAHOGANY CASE, $7\frac{3}{4} \times 5\frac{1}{4} \times 1\frac{3}{4}$, containing:—6-inch electrum pencil compass, with double knee-joints and needle point, ink point and lengthening bar—5-inch sector joint divider—one each ink and pencil needle-pointed bows—one each ink and pencil needle-pointed spring bows—two drawing pens—6-inch ivory protractor—6-inch ivory scale—two transparent set squares—and two pearwood curves, **£3.**



No. 717.

717A MAHOGANY CASE, as above, but fitted with plain steel points to compass, bows, and spring bows, and boxwood, instead of ivory, rules, **£2 8s. 6d.**

W. H. HARLING, 47 Finsbury Pavement, London, E.C.

Contractor to H.M. War Department, Admiralty, Council of India, &c.

ESTABLISHED 1851. Regd. Teleg. Address, CLINOGRAPH, LONDON.

ILLUSTRATED CATALOGUE of Drawing Instruments, Boards, Set Squares, Tee Squares, Drawing Papers, &c., Post Free on application.



A NOTABLE CATALOGUE
OF
ENGINEERS' TOOLS AND
APPLIANCES

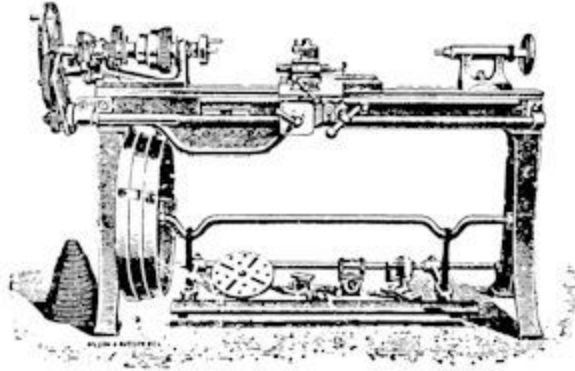


containing over
300 quarto pages
and 1600
Illustrations,
printed and bound
in superior style,
**WILL BE
GIVEN
AWAY**
on receipt of 5d.

stamps to cover postage.



**LATHES, PLANING and DRILLING MACHINES, STOCKS and
DIES, CHUCKS, DRILLS, &c., &c.**



MICROMETER GAUGES AND PRECISION TOOLS
Special Terms on application to Tool Clubs

R. MELHUISH, SONS & CO.

84-87 FETTER LANE,
HOLBORN CIRCUS, LONDON.

Transcriber's notes:

In the text version, italics are represented by underscores, and bold and black letter text by =equals= symbols. Superscripts are represented by ^{} and subscripts by _{}.

Missing or incorrect punctuation has been repaired and inconsistent spelling and hyphenation have been left.

The following mistakes have been noted:

- p. 2. orginal has been changed to original.
- p. 30. Asphalte has been changed to Asphalt.
- p. 31. Streatfeild has been changed to Streatfield (printed correctly earlier in the text).

*** END OF THE PROJECT GUTENBERG EBOOK A SHORT LIST OF
THE PUBLICATIONS OF E. & F. N. SPON, LTD. JUNE 1901 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.